

Python OOP Tasks Documentation

Programming-1 (OOP)

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Task 1: Phonebook Manager

Description: This task creates a simple phonebook system using Python dictionaries. The program allows users to add, search, and delete contacts.

Functions Used:

- **add_contact(name, phone)** → Adds a new contact to the phonebook dictionary.
- **search_contact(name)** → Searches for a contact and prints the phone number if found.
- **delete_contact(name)** → Deletes a contact from the dictionary.

Programming Concepts:

- Dictionary data structure
- Functions in Python
- Conditional statements
- Formatted output

Task 2: Student Exam Scores

Description: This task calculates the average exam score for each student stored in a dictionary.

Functions Used:

- **compute_averages(scores_dict)** → Calculates the average score for every student.

Programming Concepts:

- Dictionary comprehension
- sum() and len() functions
- Looping through dictionaries

Task 3: Find Maximum in Matrix

Description: This task searches through a 2D matrix to find the maximum value and its position.

Functions Used:

- **find_max_in_matrix(matrix)** → Finds the maximum number and returns its row and column.

Programming Concepts:

- Nested loops
- 2D lists
- Tracking indexes

Task 4: Transpose of Matrix

Description: This task creates the transpose of a matrix by swapping rows and columns.

Functions Used:

- **transpose(matrix)** → Returns the transposed matrix.

Programming Concepts:

- Matrix manipulation
- Nested loops
- List indexing

Task 5: Remove Duplicates from List

Description: This task removes duplicate values from a list while preserving the original order.

Functions Used:

- **remove_duplicates(lst)** → Returns a new list without duplicates.

Programming Concepts:

- Lists
- Membership checking
- Looping

Task 6: Factorial & Combinations

Description: This task uses recursion to calculate factorials and combinations.

Functions Used:

- **factorial(n)** → Recursively calculates factorial.
- **combination(n, r)** → Calculates combinations using factorials.

Programming Concepts:

- Recursion
- Mathematical formulas
- Base cases

Task 7: Banking System

Description: This task simulates a banking system that supports account creation, deposits, withdrawals, and balance checking.

Functions Used:

- **create_account()** → Creates a new bank account.
- **deposit()** → Adds money to the account.
- **withdraw()** → Withdraws money from the account.
- **check_balance()** → Displays account balance.

Programming Concepts:

- Global variables
- Validation
- Dictionary of dictionaries

Task 8: Event Ticket Booking

Description: This task builds a ticket booking system with discounts, taxes, and invoice generation.

Functions Used:

- **run_ticket_booking()** → Handles booking process and calculates total price.

Programming Concepts:

- Input validation
- Discount calculations
- Formatted invoices

Task 9: Restaurant Order System

Description: This task simulates a restaurant POS system with discounts, taxes, and service charges.

Functions Used:

- **run_restaurant()** → Processes restaurant orders and generates receipts.

Programming Concepts:

- Conditional discounts
- Tax calculations
- Menu management